EnergyMonitor Test Plan and Report

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Version Control

Table 1

Document version control

Version	Date	Description of change	Authors
1.0	10/14/2026	First version: inventory, traceability and results of the run without a database.	Cortés Parra, J. E.; García Artunduaga, M. A.
1.1	10/20/2026	Run of the full suite against MySQL (1131 tests, 0 failures) locally and in GitHub Actions; DEF-01 and DEF-02 closed; 3 new push notification test classes; traceability, risks and evidence updated.	Cortés Parra, J. E.; García Artunduaga, M. A.
1.2	10/24/2026	Services architecture: items under test per repository (gateway, security, device and measurement services, backend) with the `main` commits in production, totals of each suite, end-to-end tests of `energy-monitor-infra` and pipelines per repository.	Cortés Parra, J. E.; García Artunduaga, M. A.
1.3	10/28/2026	Complete migration: eight Java suites per repository (backend retired), web with 228 tests, five end-to-end tests of `energy-monitor-infra`, static verification and manual test protocol of the mobile application, traceability of the acceptance of terms (RF-SEC-13) and of the rules for Google accounts, and updated risks.	Cortés Parra, J. E.; García Artunduaga, M. A.
1.4	10/28/2026	New tests of the unusual consumption alert, the photo viewer, the photos of the members, the Settings help modals and push in Brave; updated `main` commits; traceability with RF-ALE-07, RF-HOM-12, RF-CLI-07 and RF-CLI-08 (112 requirements, 75 with an automated test).	Cortés Parra, J. E.; García Artunduaga, M. A.

Abstract

This document is the test plan and report of EnergyMonitor. It inventories the 129 automated test classes of the core service (1097 declared test methods) and the 20 suites of the web application (141 tests), traces the 107 functional and non-functional requirements of the specification against those tests, and records the run of October 14, 2026. The core service suite without a database ran 835 tests, 833 passed and 2 failed; the web ran 141 tests, all passed. The 37 classes that need MySQL were not run because the SPRING_DATASOURCE_PASSWORD variable is missing. The mobile application has no automated tests. 4 new test classes were written and 2 defects and 1 documentary observation were recorded.

Version 1.1 (October 20, 2026) records the run of the full core service suite against MySQL 8.4: 1131 tests in 132 classes, 0 failures, both locally and in the required `Tests` check of GitHub Actions. The 37 pending classes now run, defects DEF-01 and DEF-02 are closed and the web passes its 141 tests again. Traceability stands at 68 requirements with an automated test run and 39 with no coverage.

Version 1.2 (October 24, 2026) corresponds to the services architecture in production: the security, device and measurement modules left the core service for their own repositories (`en-monitor-security-service`, `en-monitor-device-service` and `en-monitor-measurement-service`), behind an API gateway (`en-monitor-api-gateway`), and the core service keeps home, alert, recommendation and notification. Each repository runs its suite in its pipeline: backend 481 tests, measurement service 151, device service 130 and gateway 57 (October 24), and security service 660 (October 22), all green, with `ModularityTests` green. Two end-to-end tests on the local

stack (`infra/e2e/device-flow.sh`, 21/21, and `infra/e2e/measurement-flow.sh`, 25/25) verify what crosses the services. The per-class inventory and the traceability of this document are still those of version 1.1.

Version 1.3 (October 28, 2026) corresponds to the complete migration: home, notification, alert and recommendation are also independent services (`en-monitor-home-service`, `en-monitor-notification-service`, `en-monitor-alert-service` and `en-monitor-recommendation-service`) and the core service was retired. Eight Java suites green (security 773, homes 260, notifications 234, measurements 151, devices 129, alerts 119, recommendations 71 and gateway 69), the web with 38 files and 228 tests, and five end-to-end tests of `energy-monitor-infra` (`device-flow` 24, `measurement-flow` 28, `alert-flow` 36, `activity-flow` 60 and `account-deletion-flow` 12). Traceability goes to 108 requirements with the acceptance of terms (RF-SEC-13): 71 with an automated test (with the two language requirements of the mobile application) and 37 without coverage. For the mobile application a unit suite of 78 tests in 8 suites was added, all passed, and type checking and `expo lint` ran without errors; a protocol of 12 manual tests was also defined, still to be run on a device.

Version 1.4 (October 28, 2026) adds tests of the latest work: the unusual consumption rule of a device (`RegisterAnomalyAlertServiceTest`, 8 cases, and its place in the readings consumer), the wording of that alert in mail and push, the profile with a photo (`UserProfileQueryServiceTest`, the home adapter and the members endpoint), the photo viewer and the photos of the members, the Settings help modals (documentation by role, support only for users and news) and the explanation of push in Brave. Totals: 1,817 tests in the eight Java services (security 774, homes 260, notifications 235, measurements 151, devices 129, alerts 128, recommendations 71 and gateway 69), 253 in the web (44 files) and 78 in the mobile. Traceability goes to 112 requirements: 75 with an automated test and 37 without coverage.

Keywords: software testing, traceability, JUnit, Vitest, Spring Modulith, EnergyMonitor

Introduction

Purpose

To document the tests that already exist and run in EnergyMonitor, and show which requirements they cover and which they do not before the delivery on October 28, 2026. The structure is adapted from IEEE 829-2008 and ISO/IEC/IEEE 29119-3.

Scope

The API gateway and the seven services (security, devices, measurements, homes, notifications, alerts and recommendations), the web application, MQTT ingestion and broker authorization are tested. Up to version 1.1 those contexts were the seven modules of the core service; since 1.2 they were split into services, the core service was retired on October 28, 2026 and each service is tested in its own repository, and the end-to-end tests of `energy-monitor-infra` verify the communication between them (gateway routes, internal API, broker and events on RabbitMQ). The device firmware, the Mosquitto broker, Bluetooth, push notifications and the Android mobile application are outside the automated tests and are declared as known gaps; for the mobile application, type checking and static analysis are also run and a manual test protocol is defined (version 1.3). Performance, load and accessibility (WCAG) are not tested; the specification sets no target for them.

Document references

Traceability uses the codes of the requirements specification (RF-SEC, RF-HOM, RF-DEV, RF-MED, RF-ALE, RF-REC, RF-NOT, RF-CLI and RNF) and the user stories HU-01 to HU-15 (Cortés Parra & García Artunduaga, 2026). IDs, class names and codes are identical in the Spanish and English versions.

Items Under Test

Table 2 lists the items under test in version 1.4, with the branch and the `main` commit at the close of October 28, 2026; version 1.3 used those of that day before the last changes (security `f57c76b`, homes `f424083`, notifications `7b10398`, alerts `f050b56` and web `698d4c9`). Version 1.2 used `main` of October 24, 2026 (gateway `c070cec`, security `334691a`, devices `f8b8526`, measurements `ee2cb81`, backend `aa9a713`, web `e59bece`, infra `eaf961b`). Version 1.1 used `main` (backend `90ae5ef`, web `2b41cb3`, infra `7324110`, mobile `dev` `4ebccfb`) and version 1.0, `dev` (backend `d03bbd3`, web `e6edc12`).

Table 2

Items under test, version, branch and commit

Item	Version	Branch	Commit	Remark
en-monitor-api-gateway	Java 21 · Spring Boot 4.0.8 · Spring Cloud Gateway	main	0b2c164	API gateway; 69 tests
en-monitor-security-service	Java 21 · Spring Boot 4.1.1	main	de52c95	Security service; 774 tests
en-monitor-device-service	Java 21 · Spring Boot 4.1.1	main	06c1bdc	Device service; 129 tests
en-monitor-measurement-service	Java 21 · Spring Boot 4.1.1	main	8a9b544	Measurement service; 151 tests
en-monitor-home-service	Java 21 · Spring Boot 4.1.1	main	6a4e396	Home service; 260 tests
en-monitor-notification-service	Java 21 · Spring Boot 4.1.1	main	ecdbc0a	Notification service; 235 tests
en-monitor-alert-service	Java 21 · Spring Boot 4.1.1	main	0e65430	Alert service; 128 tests
en-monitor-recommendation-service	Java 21 · Spring Boot 4.1.1	main	aeeeec5	Recommendation service; 71 tests
energy-monitor-backend	Java 21 · Spring Boot 4.1.1 · Spring Modulith	main	c3c6df8	Retired on 10/28/2026: its modules are the four services above; keeps the firmware (`firmware/`, version 1.3.8)
energy-monitor-web	React 19 · Vite 8	main	27ad9f0	Web application; 44 files and 253 tests
energy-monitor-mobile	Expo SDK 57 · React Native 0.86	dev	8b5edc0	78 unit tests in 8 suites; `tsc` and `expo lint` without errors. `main` is still at `894b65a`, 193 commits behind `dev`
energy-monitor-infra	Docker Compose · Mosquitto 2 · RabbitMQ 4.1	main	ceb1543	Compose, broker and scripts validated in CI; end-to-end tests `e2e/`; replaces `devices`, retired
energy-monitor-docs-final	Markdown ES/EN	dev	—	Final documentation (this document)

Test Strategy

Test levels

Table 3

Test levels and their status in the project

No.	Level	Classes/suites	Test methods	Classes run
1	Domain	21	342	21
2	Use cases and adapters without a Spring context	51	285	51
3	Web (controllers)	10	144	10

No.	Level	Classes/suites	Test methods	Classes run
4	Security	18	195	18
5	Integration with real MySQL	28	133	28
6	Architecture	1	2	1
7	Events between modules	1	1	1
8	MQTT and broker	2	16	2
9	Web frontend (Vitest)	20	141	20

All the levels in the list exist in the project. Level 9 corresponds to the web suites. The mobile application has no automated tests. In version 1.1 every core service class runs; the counts include the 3 new push notification classes (PushDeviceTest, ExpoPushSenderTest and PushDevicePersistenceTest).

Techniques and tools

White-box tests (domain and use cases with mocked ports) and black-box tests (HTTP contracts) are combined, with equivalence partitioning and boundary values (for example the power levels at 500, 1500 and 3000 W, the thresholds and the 150 seconds of silence). The new tests follow the AAA style and the project's conventions (descriptive names with @DisplayName).

Table 4

Test tools and verified versions

Tool	Version	Use
Spring Boot / spring-boot-starter-test	4.1.1	Context tests, MockMvc and @MockitoBean (never @MockBean)
JUnit Jupiter	6.0.3	Execution and parameterized tests
Mockito	5.x	Mocked ports (5.20.0 and 5.23.0 in the local Maven repository)
AssertJ	3.27.7	Assertions
Spring Modulith (spring-modulith-starter-test)	2.1.1	ModularityTests: module boundaries
Vitest / Testing Library (React)	5.0.3 / 16.3.3	Web application tests (jsdom)
Java / Node.js / Docker	21.0.12 / 22.22.2 / 29.8.2	Execution environment
JaCoCo	Not in pom.xml	No coverage report (see recommendations)
GitHub Actions	ubuntu-latest runner	Continuous integration: `Tests` (gateway, services and backend), `Tests and build` (web) and `Validate compose and scripts` (infra) checks
End-to-end scripts (`infra/e2e/`)	bash, `jq`, `openssl`, `mosquitto_pub`	Walkthroughs across services on the full local stack (version 1.2)

Test Environment

Persistence and full-context tests run against real MySQL 8.4 (container energy-monitor-mysql), not against H2. Liquibase owns the schema and Hibernate is set to ddl-auto: validate; an H2 database or @AutoConfigureTestDatabase(replace = ANY) would change the dialect and defeat the validation of the real schema. The cost is that these tests are not hermetic: they need MySQL running, the test profile and the SPRING_DATASOURCE_PASSWORD variable; the default user is energy_app.

Table 5

Environment components

Component	Detail
MySQL 8.4	Docker, energy-monitor-mysql (healthy)
Spring profile	test (src/test/resources/application-test.yaml); recommendation.enabled: false
CI (GitHub Actions)	`mysql:8.4` service container (user energy_app), `application-local.yaml` built from the example and a throwaway RSA pair; the `Tests` job takes about 23 minutes

Entry, Exit, Suspension and Resumption Criteria

Table 6

Test criteria

Criterion	Condition
Entry	Compilable code on dev; MySQL running and SPRING_DATASOURCE_PASSWORD defined for context tests; npm install run on the web.
Exit	All automated tests passed, no open high-severity defect and the classes with MySQL run.
Suspension	MySQL unavailable, missing credentials or a build failure.
Resumption	Environment restored; resume from the class that stopped.

Inventory of Automated Tests

Each table corresponds to a module of the core service as it was in version 1.1, when the seven modules lived in `energy-monitor-backend`. Since version 1.2 the code of security, device and measurement is in their own repositories, which have their own suites (Table 2); this inventory was not redone per class for version 1.2. “Status” is Passed, Fails (with its defect) or “To run (MySQL)” for the classes that boot a context with a database. New classes are marked NEW in their description. In version 1.1 every class ran against MySQL and passed, so the status of all of them is “Passed”.

Table 7

Tests of the security module

Level	Test class	What it verifies	Methods	Requirements covered	Status
Domain	AuditLogTest	Rules and invariants of AuditLog	9	RF-SEC-12	Passed
Domain	PasswordResetTokenTest	Rules and invariants of PasswordResetToken	19	RF-SEC-06	Passed
Domain	RbacTest	Rules and invariants of Rbac	14	RF-SEC-09	Passed
Domain	RefreshTokenTest	Rules and invariants of RefreshToken	41	RF-SEC-04, RNF-SEG-02	Passed
Domain	SecuritySupportModelTest	Rules and invariants of SecuritySupportModel	28	RF-SEC-12	Passed
Domain	UserSessionTest	Rules and invariants of UserSession	45	RF-SEC-04	Passed
Domain	UserTest	Rules and invariants of User	21	RF-SEC-01	Passed
Domain	ValueObjectTest	Rules and invariants of ValueObject	14	RF-SEC-01	Passed

Level	Test class	What it verifies	Methods	Requirements covered	Status
Use cases and adapters	AccountLifecycleServiceTest	Account query, profile edit and account state change	9	RF-SEC-08	Passed
Use cases and adapters	AuthenticateUserServiceTest	Account state and credentials classified; every failure is recorded	6	RF-SEC-03	Passed
Use cases and adapters	ChangePasswordServiceTest	Password change with current-password check and policy	3	RF-SEC-07	Passed
Use cases and adapters	DeleteAccountServiceTest	Deleting the account checks the password and announces AccountDeleted	2	RF-SEC-11	Passed
Use cases and adapters	EmailVerificationServiceTest	Six-digit verification code valid for at least 15 minutes	6	RF-SEC-02	Passed
Use cases and adapters	PasswordResetServiceTest	Recovery: single-use code, validity window and attempt budget	16	RF-SEC-06	Passed
Use cases and adapters	RefreshSessionServiceTest	Renewal: single use, lineage, session identity and replay escalation	20	RF-SEC-04, RNF-SEG-01, RNF-SEG-02	Passed
Use cases and adapters	RegisterUserServiceTest	Registration: password hash, person and user, duplicate and policy rejection	7	RF-SEC-01, RF-SEC-12	Passed
Use cases and adapters	SignInWithGoogleServiceTest	Sign in with Google and account creation	7	RF-SEC-05, RNF-INT-03	Passed
Use cases and adapters	UserAdministrationServiceTest	Account administration: an administrator never acts on their own account	18	—	Passed
Use cases and adapters	UserProfileQueryServiceTest	Name and email of known accounts	1	RF-SEC-08	Passed
Use cases and adapters	UserSessionServiceTest	Session opening with opaque refresh token and logout	6	RF-SEC-04	Passed
Web (controllers)	AdminControllerWebTest	Administration endpoints /api/v1/admin with RBAC permissions and 409 on self-administration	15	—	Passed
Web (controllers)	AuthControllerWebTest	HTTP contract of authentication, account, password, profile, Google and permissions (401, 403, 429)	66	RF-SEC-01, RF-SEC-02, RF-SEC-03, RF-SEC-05, RF-SEC-07, RF-SEC-08, RF-SEC-09, RNF-SEG-06	Passed
Security	AuthRefreshEndpointWebTest	HTTP contract of POST /api/v1/auth/refresh	18	RF-SEC-04	Passed
Security	BCryptPasswordHasherAdapterTest	BCryptPasswordHasherAdapter	10	RF-SEC-03	Passed
Security	InMemoryResetAttemptLimiterTest	Limit of 10 attempts per IP every 15 minutes	8	RNF-SEG-06	Passed
Security	JwtTokenIssuerTest	RS256 JWT issuing and validation (15 minutes)	18	RF-SEC-03, RNF-SEG-01	Passed

Level	Test class	What it verifies	Methods	Requirements covered	Status
Security	PemRsaKeysTest	Which PEM keys are accepted and the message on refusal	19	—	Passed
Security	PepperedHmacResetCodeHasherTest	PepperedHmacResetCodeHasher	15	RF-SEC-06, RNF-SEG-05	Passed
Security	PoolSaturatedLimiterTest	The per-IP limit applies without waiting for a pool connection (429)	2	RNF-SEG-06	Passed
Security	RbacServiceTest	Assign and revoke roles with audit and permission checks	17	RF-SEC-09	Passed
Security	ResetAttemptLimiterBeforeDatabaseTest	A redemption refused by the limit opens no transaction	3	RNF-SEG-06	Passed
Security	SecureIdentifierGeneratorAdapterTest	SecureIdentifierGeneratorAdapter	5	—	Passed
Security	SecureTokenGeneratorAdapterTest	SecureTokenGeneratorAdapter	13	—	Passed
Security	SecurityCorsConfigurationTest	CORS policy guards without booting a context	11	RNF-SEG-03	Passed
Security	SecurityCorsWebTest	CORS contract: origins answered and preflight response	23	RNF-SEG-03	Passed
Security	SecurityLoginFlowTest	Order of the login exchange and what it refuses	11	RF-SEC-03	Passed
Security	Sha256TokenHasherTest	Sha256TokenHasher	9	RF-SEC-04	Passed
Integration with real	ConfigurationPolicyPersistenceTest	ConfigurationPolicy round trips against real MySQL	6	RF-SEC-12	Passed
Integration with real	LogPersistenceTest	Log round trips against real MySQL	5	RF-SEC-12	Passed
Integration with real	PasswordPolicySeedTest	Password policy seeded by changeset security-019	5	RF-SEC-12	Passed
Integration with real	PasswordResetTokenReservationTest	Redemption attempt reservation against MySQL	6	RF-SEC-06	Passed
Integration with real	RbacAssignmentPersistenceTest	RbacAssignment round trips against real MySQL	2	RF-SEC-09	Passed
Integration with real	RefreshLifecycleIntegrationTest	Full lifecycle: open session, rotate, detect replay and revoke the family	15	RF-SEC-04, RNF-SEG-02	Passed
Integration with real	RefreshSessionTransactionTest	Refresh token rotation is atomic against the real database	6	RF-SEC-04	Passed
Integration with real	RefreshTokenPersistenceTest	RefreshToken round trips against real MySQL	10	RF-SEC-04	Passed
Integration with real	SecurityTransactionBoundaryTest	Where the transaction boundary sits and that Spring honours it	5	—	Passed
Integration with real	SecurityTransactionRollbackTest	The transaction really rolls back	3	—	Passed
Integration with real	TokenSessionPersistenceTest	TokenSession round trips against real MySQL	6	RF-SEC-04	Passed

Level	Test class	What it verifies	Methods	Requirements covered	Status
Integration with real	UserPersonPersistenceTest	UserPerson round trips against real MySQL	5	RF-SEC-01, RNF-MAN-03	Passed

Table 8

Tests of the home module

Level	Test class	What it verifies	Methods	Requirements covered	Status
Domain	HomeTest	Rules and invariants of Home	16	RF-HOM-01	Passed
Domain	HomeThresholdsTest	Rules and invariants of HomeThresholds	17	RF-HOM-08	Passed
Domain	HomeTypeTest	Rules and invariants of HomeType	8	—	Passed
Domain	MemberPermissionsTest	Rules and invariants of MemberPermissions	3	RF-HOM-05	Passed
Domain	OwnerPermissionsTest	Rules and invariants of OwnerPermissions	3	RF-HOM-05	Passed
Domain	RoleTest	Rules and invariants of Role	3	RF-HOM-05	Passed
Domain	UserHomeTest	Rules and invariants of UserHome	26	RF-HOM-05	Passed
Use cases and adapters	CreateHomeServiceTest	CreateHome use case: business rules and errors with mocked ports	6	RF-HOM-01	Passed
Use cases and adapters	DeleteHomeServiceTest	Only the owner deletes the household; announces HomeDeleted; a member gets 403	4	RF-HOM-09, RNF-MAN-03	Passed
Use cases and adapters	GetHomeThresholdsServiceTest	GetHomeThresholds use case: business rules and errors with mocked ports	2	RF-HOM-08	Passed
Use cases and adapters	GetHomeTypeServiceTest	GetHomeType use case: business rules and errors with mocked ports	2	—	Passed
Use cases and adapters	JoinHomeServiceTest	JoinHome use case: business rules and errors with mocked ports	4	RF-HOM-03	Passed
Use cases and adapters	LeaveHomeServiceTest	LeaveHome use case: business rules and errors with mocked ports	4	RF-HOM-06	Passed
Use cases and adapters	ListHomeTypesServiceTest	ListHomeTypes use case: business rules and errors with mocked ports	2	—	Passed
Use cases and adapters	ListHomesServiceTest	ListHomes use case: business rules and errors with mocked ports	5	RF-HOM-02	Passed
Use cases and adapters	ListMembersServiceTest	ListMembers use case: business rules and errors with mocked ports	4	RF-HOM-04	Passed
Use cases and adapters	RemoveDeletedAccountServiceTest	Deleted account: drops owned households and refuses if sole owner with members	2	RF-SEC-11	Passed

Level	Test class	What it verifies	Methods	Requirements covered	Status
Use cases and adapters	RemoveUserServiceTest	RemoveUser use case: business rules and errors with mocked ports	5	RF-HOM-06	Passed
Use cases and adapters	ToggleFavoriteServiceTest	ToggleFavorite use case: business rules and errors with mocked ports	2	RF-HOM-07	Passed
Use cases and adapters	TransferHomeOwnershipServiceTest	Ownership transfer: roles swapped, single owner, invalid targets	6	RF-HOM-10	Passed
Use cases and adapters	UpdateHomeThresholdsServiceTest	UpdateHomeThresholds use case: business rules and errors with mocked ports	7	RF-HOM-08	Passed
Web (controllers)	HomeControllerTest	Household endpoints: create, list, join, favorite, remove, leave, delete and transfer	26	RF-HOM-01, RF-HOM-02, RF-HOM-03, RF-HOM-04, RF-HOM-06, RF-HOM-07, RF-HOM-08, RF-HOM-09, …	Passed
Web (controllers)	HomeExceptionHandlerTest	What the module answers when a request cannot be attributed to a user	2	RF-SEC-10	Passed
Integration with real	HomePersistenceTest	Home round trips against real MySQL	6	RF-HOM-01, RNF-MAN-03	Passed
Integration with real	HomeThresholdsPersistenceTest	HomeThresholds round trips against real MySQL	3	RF-HOM-08	Passed
Integration with real	HomeTypePersistenceTest	HomeType round trips against real MySQL	3	—	Passed
Integration with real	HomeTypeSeedTest	The catalog holds exactly the six seeded household types	1	RF-HOM-01	Passed
Integration with real	UserHomePersistenceTest	UserHome round trips against real MySQL	7	RF-HOM-05	Passed

Table 9

Tests of the device module

Level	Test class	What it verifies	Methods	Requirements covered	Status
Use cases and adapters	BrokerAccessServiceTest	Broker authentication of the clear key against the hash and topic authorization	8	RF-DEV-07, RNF-SEG-09	Passed
Use cases and adapters	DeviceConnectivityServiceTest	Online and offline status, change log and 150 s of silence	6	RF-ALE-02, RF-MED-05, RF-MED-06, RNF-DIS-02	Passed
Use cases and adapters	HomeDeviceServiceTest	Link, list, edit, reissue credentials and unlink with owner rules	9	RF-DEV-02, RF-DEV-03, RF-DEV-04, RF-DEV-05, RF-DEV-06	Passed
Web (controllers)	HomeDeviceControllerTest	NEW. Household device endpoints: link, edit, credentials, unlink and 400/403/404/409 errors	10	RF-DEV-02, RF-DEV-03, RF-DEV-04, RF-DEV-05, RF-DEV-06	Passed
Security	ApiKeyHashTest	SHA-256 hash of the device api_key	3	RF-DEV-07, RNF-SEG-09	Passed
MQTT and broker	BrokerAuthControllerTest	NEW. /internal/mqtt/auth/user and /acl: 200 grants, 403 denies, access codes 1 to 4	8	RF-DEV-07	Passed

Table 10

Tests of the measurement module

Level	Test class	What it verifies	Methods	Requirements covered	Status
Domain	ConsumptionLevelTest	LOW, MEDIUM, HIGH and CRITICAL levels and their limits	14	RF-MED-09	Passed
Domain	MeasurementTest	Rules and invariants of Measurement	12	RF-MED-02	Passed
Use cases and adapters	GetConsumptionStatisticsServiceTest	GetConsumptionStatistics use case: business rules and errors with mocked ports	5	RF-MED-07	Passed
Use cases and adapters	GetLatestMeasurementServiceTest	GetLatestMeasurement use case: business rules and errors with mocked ports	2	RF-MED-07	Passed
Use cases and adapters	HomeConsumptionUseCaseTest	Household consumption summary and history	8	RF-MED-08	Passed
Use cases and adapters	ListConsumptionLevelsServiceTest	ListConsumptionLevels use case: business rules and errors with mocked ports	2	RF-MED-09	Passed
Use cases and adapters	ListMeasurementsServiceTest	ListMeasurements use case: business rules and errors with mocked ports	4	RF-MED-07	Passed
Use cases and adapters	RegisterMeasurementServiceTest	RegisterMeasurement use case: business rules and errors with mocked ports	4	RF-MED-02	Passed
Web (controllers)	ConsumptionLevelControllerTest	HTTP codes and body of ConsumptionLevelController	1	RF-MED-09	Passed
Web (controllers)	MeasurementControllerTest	HTTP codes and body of MeasurementController	5	RF-MED-07	Passed
Integration with real	ConsumptionLevelPersistenceTest	ConsumptionLevel round trips against real MySQL	4	RF-MED-09	Passed
Integration with real	MeasurementPersistenceTest	Measurement round trips against real MySQL	4	RF-MED-02	Passed
MQTT and broker	TelemetryMqttConsumerTest	NEW. Telemetry ingestion: the topic identifies the device and invalid messages are dropped	8	RF-DEV-07, RF-MED-02, RF-MED-03	Passed

Table 11

Tests of the alert module

Level	Test class	What it verifies	Methods	Requirements covered	Status
Domain	AlertTest	Rules and invariants of Alert	16	RF-ALE-03, RF-ALE-04	Passed
Use cases and adapters	GetAlertServiceTest	GetAlert use case: business rules and errors with mocked ports	2	RF-ALE-05	Passed
Use cases and adapters	ListAlertsServiceTest	ListAlerts use case: business rules and errors with mocked ports	3	RF-ALE-05	Passed
Use cases and adapters	RegisterDeviceLinkedAlertServiceTest	Device linked notice on the first reading	4	RF-ALE-01	Passed
Use cases and adapters	RegisterLimitAlertServiceTest	Daily or monthly limit alerts	4	RF-ALE-04	Passed
Use cases and adapters	RegisterThresholdAlertServiceTest	High or critical consumption alerts by power level	7	RF-ALE-03	Passed
Use cases and adapters	ResolveAlertServiceTest	ResolveAlert use case: business rules and errors with mocked ports	4	RF-ALE-05	Passed
Web (controllers)	AlertControllerTest	Alert endpoints: list, get, read, delete and 404	7	RF-ALE-05	Passed
Integration with real	AlertPersistenceTest	Alert round trips against real MySQL	4	RF-ALE-05	Passed

Level	Test class	What it verifies	Methods	Requirements covered	Status
Events between modules	MeasurementRecordedListenerTest	The MeasurementRecorded event triggers threshold and limit alert registration	1	RF-ALE-03, RF-ALE-06	Passed

Table 12

Tests of the recommendation module

Level	Test class	What it verifies	Methods	Requirements covered	Status
Domain	RecommendationRulesTest	Five saving rules: peak hours, standby, weekly comparison, projection and per-device increase	8	RF-REC-01, RF-REC-02, RF-REC-03, RF-REC-04, RF-REC-05, RF-REC-06, RF-REC-07, RF-REC-09	Passed
Use cases and adapters	EvaluateRecommendationsServiceTest	Rule evaluation per household and no repeats within 7 days	4	RF-REC-01, RF-REC-07	Passed
Use cases and adapters	RecommendationInboxServiceTest	RecommendationInbox use case: business rules and errors with mocked ports	4	RF-REC-08	Passed
Web (controllers)	RecommendationControllerTest	NEW. Recommendation inbox: list, read, delete, 404 and 409	7	RF-REC-08	Passed
Integration with real	RecommendationPersistenceTest	Recommendation round trips against real MySQL	3	RF-REC-08	Passed

Table 13

Tests of the notification module

Level	Test class	What it verifies	Methods	Requirements covered	Status
Domain	NotificationTest	Rules and invariants of Notification	20	RF-NOT-01, RF-NOT-05	Passed
Use cases and adapters	AbandonedDeliverySweeperTest	Sweep of abandoned deliveries at startup	4	RF-NOT-05	Passed
Use cases and adapters	AdminNotificationsTest	Administrator notifications: who receives each event and reads scoped to the caller	5	—	Passed
Use cases and adapters	EmailPasswordResetDeliveryAdapterTest	Recovery code delivery; nothing is sent if the transaction rolls back	5	RF-NOT-06, RF-SEC-06	Passed
Use cases and adapters	EmailSenderTest	Recording of the attempt before sending and sending the email	9	RF-NOT-01, RF-NOT-05, RF-NOT-06, RF-SEC-02, RNF-INT-02	Passed
Use cases and adapters	NotifyAlertTest	Delivery of alerts and recommendations to members by email and push per preferences	8	RF-NOT-01, RF-NOT-02, RF-NOT-07	Passed
Use cases and adapters	SendPasswordResetEmailTest	Password recovery message and its record	11	RF-NOT-06, RF-SEC-06	Passed
Use cases and adapters	WebPushSenderTest	Push message encryption (RFC 8291) as a browser decrypts it	2	RF-NOT-01, RF-NOT-04	Passed
Domain	PushDeviceTest	NEW (1.1). Invariants of the mobile device registered for push	5	RF-NOT-01	Passed
Use cases and adapters	ExpoPushSenderTest	NEW (1.1). Interpretation of the Expo push relay's answer	9	RF-NOT-01	Passed

Level	Test class	What it verifies	Methods	Requirements covered	Status
Integration with real	PushDevicePersistenceTest	NEW (1.1). Registration and removal of mobile installations against the real schema	7	RF-NOT-01	Passed
Web (controllers)	AdminNotificationsControllerWebTest	Administrator notification endpoints	5	—	Passed
Integration with real	NotificationPersistenceTest	Delivery log against the real schema	4	RF-NOT-05	Passed

Table 14

Tests of the cross-module module

Level	Test class	What it verifies	Methods	Requirements covered	Status
Security	AdminRbacIntegrationTest	Seeded ADMIN role, USER.READ, USER.MANAGE and ROLE.MANAGE codes and directory searches	6	RF-SEC-09	Passed
Security	HomeCallerIdentityIntegrationTest	Caller identity on /homes through the real filter chain and an RS256 JWT; X-User-Id is ignored	4	RF-SEC-10	Passed
Integration with real	AccountDeletionIntegrationTest	Account deletion across security and home against the real schema	3	RF-SEC-11	Passed
Integration with real	DeviceApiKeyStorageIntegrationTest	Only the SHA-256 hash of the api_key reaches the table; broker login keeps working	3	RF-DEV-02, RNF-SEG-09	Passed
Integration with real	EnergyMonitorBackendApplicationTests	Boot of the full application context	1	—	Passed
Integration with real	HomeDeletionIntegrationTest	Deleting a household drops memberships and thresholds and unlinks devices; a member cannot	2	RF-DEV-06, RF-HOM-09	Passed
Integration with real	HomeOwnershipTransferIntegrationTest	Atomic swap of OWNER and MEMBER roles against the real schema	4	RF-HOM-05, RF-HOM-10	Passed
Architecture	ModularityTests	ApplicationModules.verify(): the boundaries of the seven modules hold	2	RNF-MAN-01	Passed

Table 15

Web application tests (Vitest)

Suite	Tests	Requirements covered	Status
components/layout/MainLayout/Sidebar/Sidebar.test.jsx	2	RF-CLI-01	Passed
context/HomeContext.test.jsx	11	—	Passed
features/Account/Account.test.jsx	6	RF-CLI-01, RF-SEC-08, RF-SEC-11	Passed
features/Account/AccountDeleteError.test.jsx	1	RF-SEC-11	Passed
features/Admin/AdminNotifications.test.jsx	5	—	Passed
features/Admin/AdminUsers.test.jsx	9	—	Passed
features/DetailHome/Home/DeleteHome/DeleteHome.test.jsx	5	RF-CLI-01, RF-HOM-09	Passed
features/DetailHome/Users/Users.test.jsx	12	RF-CLI-01, RF-HOM-04, RF-HOM-06, RF-HOM-10	Passed
features/auth/components/RPasswordForm/recover.test.jsx	4	RF-CLI-01	Passed
features/auth/components/RegisterForm/verify.test.jsx	4	RF-CLI-01	Passed
features/dashboard/pages/DashboardPage.test.jsx	5	RF-CLI-01, RF-HOM-02	Passed
features/shared/homeTypes.test.js	6	—	Passed
routes/RequirePermission.test.jsx	3	—	Passed
routes/UserOnlyRoute.test.jsx	4	RF-CLI-01	Passed
services/admin/adminApi.test.js	4	—	Passed

Suite	Tests	Requirements covered	Status
services/auth/authApi.test.js	4	—	Passed
services/home/homeApi.test.js	15	—	Passed
services/http/errorMessages.test.js	27	—	Passed
services/http/httpClient.refresh.test.js	3	—	Passed
services/http/httpClient.test.js	11	—	Passed

Automated tests of the mobile application do not exist: the repository has no test files or test script (only expo lint). It is a declared gap. Since version 1.3 type checking (`tsc`) and `expo lint` are run, and a manual test protocol is defined (Table 38).

Requirement → Test Traceability Matrix

The 107 requirements of the specification are crossed with the test classes. “Coverage” distinguishes what was run from what depends on MySQL. Result in version 1.0: 66 with an automated test run, 1 with automated tests that depend on MySQL and 40 with no coverage. Result in version 1.1: 68 with an automated test run (RNF-MAN-01 now runs and RF-NOT-02 is covered by NotifyAlertTest) and 39 with no coverage.

Table 16

Traceability of the RF-SEC requirements

Requirement	Description	Automated tests	Coverage
RF-SEC-01	The system must allow requesting the registration of an account with personal data, email and password…	RegisterUserServiceTest, MistypedEmailDomainsTest, AuthControllerWebTest, UserPersonPersistenceTest, UserTest, ValueObjectTest	Automated
RF-SEC-02	The system must send a six-digit verification code to the given email and create the account…	EmailVerificationServiceTest, EmailSenderTest, AuthControllerWebTest	Automated
RF-SEC-03	The system must allow signing in securely with email and password	AuthenticateUserServiceTest, SecurityLoginFlowTest, AuthControllerWebTest, JwtTokenIssuerTest, BCryptPasswordHasherAdapterTest	Automated
RF-SEC-04	The system must keep the user's session without asking for credentials continuously, let the user sign out…	RefreshSessionServiceTest, UserSessionServiceTest, UserSessionTest, RefreshTokenTest, AuthRefreshEndpointWebTest, RefreshLifecycleIntegrationTest, …	Automated
RF-SEC-05	The system must allow registering and signing in with a Google account, with acceptance of the terms	SignInWithGoogleServiceTest, AuthControllerWebTest	Automated
RF-SEC-06	The system must allow password recovery with a six-digit code sent to the email, valid for 15 minutes and…	PasswordResetServiceTest, PasswordResetTokenTest, PepperedHmacResetCodeHasherTest, SendPasswordResetEmailTest, EmailPasswordResetDeliveryAdapterTest, PasswordResetTokenReservationTest, …	Automated
RF-SEC-07	The system must allow a signed-in user to change their password (not Google accounts)	ChangePasswordServiceTest, AuthControllerWebTest, account-deletion-flow.sh	Automated
RF-SEC-08	The system must allow viewing the account data and updating the profile (first and last name separately)	AccountLifecycleServiceTest, UserProfileQueryServiceTest, AuthControllerWebTest, Account.test.jsx	Automated
RF-SEC-09	The system must allow knowing the user's permissions	RbacServiceTest, RbacTest, RbacAssignmentPersistenceTest, AdminRbacIntegrationTest, AuthControllerWebTest	Automated
RF-SEC-10	The system must identify the user only from their session and must not accept identities indicated by the…	HomeCallerIdentityIntegrationTest, HomeExceptionHandlerTest	Automated

Requirement	Description	Automated tests	Coverage
RF-SEC-11	The system must allow the user to delete their account by confirming their password or, with Google, a code	DeleteAccountServiceTest, RemoveDeletedAccountServiceTest, AccountDeletionIntegrationTest, AccountDeleteError.test.jsx, Account.test.jsx, account-deletion-flow.sh	Automated
RF-SEC-12	The system must apply a password policy (minimum length of 8 and maximum of 64 characters) and…	PasswordPolicySeedTest, ConfigurationPolicyPersistenceTest, LogPersistenceTest, AuditLogTest, SecuritySupportModelTest, RegisterUserServiceTest	Automated
RF-SEC-13	The system must not allow access to an account that has not accepted the terms and conditions…	RegisterUserServiceTest, SignInWithGoogleServiceTest, AuthControllerWebTest, SecurityLoginFlowTest, UserPersonPersistenceTest, registerSchema.test.ts, registerPayload.test.ts, termsConsent.test.ts (mobile)	Automated

Table 17

Traceability of the RF-HOM requirements

Requirement	Description	Automated tests	Coverage
RF-HOM-01	The system must allow creating a home by choosing a home type from a catalog	CreateHomeServiceTest, HomeControllerTest, HomeTest, HomePersistenceTest, HomeTypeSeedTest	Automated
RF-HOM-02	The system must show the user the homes they belong to	ListHomesServiceTest, HomeControllerTest, DashboardPage.test.jsx	Automated
RF-HOM-03	The system must allow joining a home with its access code, with the member role	JoinHomeServiceTest, HomeControllerTest	Automated
RF-HOM-04	The system must show the members of a home	ListMembersServiceTest, HomeControllerTest, Users.test.jsx	Automated
RF-HOM-05	The system must distinguish users' general permissions from the owner and member roles within each home	UserHomeTest, OwnerPermissionsTest, MemberPermissionsTest, RoleTest, UserHomePersistenceTest, HomeOwnershipTransferIntegrationTest	Automated
RF-HOM-06	The owner must be able to remove a member and any member must be able to leave the home	RemoveUserServiceTest, LeaveHomeServiceTest, HomeControllerTest, Users.test.jsx	Automated
RF-HOM-07	The user must be able to mark a home as a favorite	ToggleFavoriteServiceTest, HomeControllerTest	Automated
RF-HOM-08	The system must allow viewing, updating and resetting to system values the thresholds of a home	GetHomeThresholdsServiceTest, UpdateHomeThresholdsServiceTest, HomeThresholdsTest, HomeThresholdsPersistenceTest, HomeControllerTest	Automated
RF-HOM-09	The owner must be able to delete the home by soft delete	DeleteHomeServiceTest, HomeDeletionIntegrationTest, HomeControllerTest, DeleteHome.test.jsx	Automated
RF-HOM-10	The owner must be able to transfer home ownership to another member	TransferHomeOwnershipServiceTest, HomeOwnershipTransferIntegrationTest, HomeControllerTest, Users.test.jsx	Automated
RF-HOM-11	Pending: limit a device's history to the linking period	—	No coverage
RF-HOM-12	The system must show the members of a home the profile photo of each one and allow enlarging it	UserProfileQueryServiceTest, HttpUserDirectoryAdapterTest, HomeControllerTest, Users.test.jsx	Automated

Table 18

Traceability of the RF-DEV requirements

Requirement	Description	Automated tests	Coverage
RF-DEV-01	The system must provide a catalog of appliance types	—	No coverage

Requirement	Description	Automated tests	Coverage
RF-DEV-02	The owner must be able to link a measuring device to their home	HomeDeviceServiceTest, HomeDeviceControllerTest, DeviceApiKeyStorageIntegrationTest	Automated
RF-DEV-03	Members must be able to see the devices of a home	HomeDeviceServiceTest, HomeDeviceControllerTest	Automated
RF-DEV-04	The owner must be able to edit the appliance, name and room of a device without intervening on the…	HomeDeviceServiceTest, HomeDeviceControllerTest	Automated
RF-DEV-05	The owner must be able to change the Wi-Fi network of an already linked device	HomeDeviceServiceTest, HomeDeviceControllerTest	Automated
RF-DEV-06	The owner must be able to unlink a device	HomeDeviceServiceTest, HomeDeviceControllerTest, HomeDeletionIntegrationTest	Automated
RF-DEV-07	The system must only accept data from linked devices, and each device can only send its own data	BrokerAccessServiceTest, BrokerAuthControllerTest, ApiKeyHashTest, TelemetryMqttConsumerTest	Automated
RF-DEV-08	Linking must be possible from the web application over Bluetooth: the application shows the Wi-Fi networks…	—	No coverage
RF-DEV-09	The device must reconnect without user intervention after a Wi-Fi loss or a temporary system failure, keep…	—	No coverage

Table 19

Traceability of the RF-MED requirements

Requirement	Description	Automated tests	Coverage
RF-MED-01	The device must send every 60 seconds the voltage (V), current (A), active power (W) and accumulated…	—	No coverage
RF-MED-02	The system must store each reading with the date and time it was taken	RegisterMeasurementServiceTest, MeasurementTest, MeasurementPersistenceTest, TelemetryMqttConsumerTest	Automated
RF-MED-03	The system must drop messages that claim to belong to a device other than the one sending them	TelemetryMqttConsumerTest	Automated
RF-MED-04	The device must report its status every five minutes and on connect: whether it is online or offline, its…	—	No coverage
RF-MED-05	The system must keep each device's state changes	DeviceConnectivityServiceTest	Automated
RF-MED-06	The system must mark a device as offline when the device announces it is disconnecting or when 150 seconds…	DeviceConnectivityServiceTest	Automated
RF-MED-07	The system must allow querying measurements, the latest measurement and consumption statistics	ListMeasurementsServiceTest, GetLatestMeasurementServiceTest, GetConsumptionStatisticsServiceTest, MeasurementControllerTest	Automated
RF-MED-08	The system must provide the consumption summary and history of a home, with date range selection	HomeConsumptionUseCaseTest	Automated

Requirement	Description	Automated tests	Coverage
RF-MED-09	The system must classify power into the levels LOW (0 to 500 W), MEDIUM (500 to 1500 W), HIGH (1500 to…	ConsumptionLevelTest, ListConsumptionLevelsServiceTest, ConsumptionLevelControllerTest, ConsumptionLevelPersistenceTest	Automated

Table 20

Traceability of the RF-ALE requirements

Requirement	Description	Automated tests	Coverage
RF-ALE-01	The system must raise a device linked notice with the first reading a device sends after being linked,…	RegisterDeviceLinkedAlertServiceTest	Automated
RF-ALE-02	The system must raise a disconnection alert when a device announces it is disconnecting or 150 seconds…	DeviceConnectivityServiceTest	Automated
RF-ALE-03	The system must raise a high or critical consumption alert when a reading falls in the HIGH or CRITICAL level	RegisterThresholdAlertServiceTest, AlertTest, MeasurementRecordedListenerTest	Automated
RF-ALE-04	The system must raise a daily or monthly limit alert when the home's energy since the start of the day or…	RegisterLimitAlertServiceTest, AlertTest	Automated
RF-ALE-05	The system must allow listing alerts, viewing the detail of one, marking it as read and deleting it	ListAlertsServiceTest, GetAlertServiceTest, ResolveAlertServiceTest, AlertControllerTest, AlertPersistenceTest	Automated
RF-ALE-06	The system must raise consumption alerts from each new reading and disconnection alerts from the loss of…	MeasurementRecordedListenerTest	Automated
RF-ALE-07	The system must raise an unusual consumption alert when a reading is at least 200 W and 2.5 times the usual power of the device…	RegisterAnomalyAlertServiceTest, MeasurementRecordedConsumerTest, AlertMessagesAnomalyTest	Automated

Table 21

Traceability of the RF-REC requirements

Requirement	Description	Automated tests	Coverage
RF-REC-01	The system must evaluate every hour the hourly energy of homes whose devices reported in the last week,…	EvaluateRecommendationsServiceTest, RecommendationRulesTest	Automated
RF-REC-02	Peak hours rule: at least 40% of the last 7 days' energy (minimum 2 kWh) was used between 18:00 and 22:00	RecommendationRulesTest	Automated
RF-REC-03	Standby consumption rule: a device that is not a refrigerator drew an average of 15 W or more between…	RecommendationRulesTest	Automated
RF-REC-04	Weekly comparison rule: the last 7 days are 25% or more above the previous 7 (minimum 1 kWh)	RecommendationRulesTest	Automated
RF-REC-05	Limit projection rule: at the current pace, the home will exceed its daily or monthly limit, whichever the…	RecommendationRulesTest	Automated
RF-REC-06	Device increase rule: a device's mean power in the last 7 days is 30% or more above that of the previous…	RecommendationRulesTest	Automated
RF-REC-07	The same recommendation (home, type and device) must not repeat within 7 days, even if it was deleted	EvaluateRecommendationsServiceTest, RecommendationRulesTest	Automated
RF-REC-08	The system must allow listing a home's recommendations, marking them as read and deleting the read ones,…	RecommendationInboxServiceTest, RecommendationControllerTest, RecommendationPersistenceTest	Automated

Requirement	Description	Automated tests	Coverage
RF-REC-09	The rules need history: peak hours and standby consumption work after a few days of readings, the weekly…	RecommendationRulesTest	Automated

Table 22

Traceability of the RF-NOT requirements

Requirement	Description	Automated tests	Coverage
RF-NOT-01	The system must deliver every alert to all members of the home by email and browser notification,…	NotifyAlertTest, NotificationTest, WebPushSenderTest, EmailSenderTest, ExpoPushSenderTest, PushDeviceTest	Automated
RF-NOT-02	The system must deliver every new recommendation through the same channels	NotifyAlertTest	Automated
RF-NOT-03	The user must be able to view and change their notification preferences	—	No coverage
RF-NOT-04	The user must be able to turn browser notifications on and off	WebPushSenderTest	Automated
RF-NOT-05	Every delivery attempt must be recorded as sent or failed, without storing the message content	NotificationTest, NotificationPersistenceTest, AbandonedDeliverySweeperTest, EmailSenderTest	Automated
RF-NOT-06	The system must send the verification code and the password recovery code by email	SendPasswordResetEmailTest, EmailPasswordResetDeliveryAdapterTest, EmailSenderTest	Automated
RF-NOT-07	Every alert email must include a link to the web application	NotifyAlertTest	Automated

Table 23

Traceability of the RF-CLI requirements

Requirement	Description	Automated tests	Coverage
RF-CLI-01	The web application must offer registration, sign-in (including Google), password recovery, account,…	Account.test.jsx, DashboardPage.test.jsx, DeleteHome.test.jsx, Users.test.jsx, verify.test.jsx, recover.test.jsx, …	Automated
RF-CLI-02	The web application must show a badge with pending alerts, a counter in the tab title and a pop-up notice…	—	No coverage
RF-CLI-03	The web application must link devices over Bluetooth and allow browser notifications, which arrive even…	—	No coverage
RF-CLI-04	The mobile application must be available for Android only and allow access to the account, homes, alerts,…	—	No coverage
RF-CLI-05	Both applications must allow choosing the language among Spanish, English, French and Portuguese	locales.test.ts (mobile)	Automated
RF-CLI-06	The applications must allow choosing a color palette with dark mode	—	No coverage
RF-CLI-07	The web application must offer help in Settings in modals: documentation by role, support only for users and news	HelpCenter.test.jsx, DocsModal.test.jsx, NewsModal.test.jsx, Support.test.jsx	Automated
RF-CLI-08	The web application must allow viewing the profile photo enlarged inside the page itself	PhotoViewer.test.jsx, Account.test.jsx	Automated

Table 24

Traceability of the RNF-SEG requirements

Requirement	Description	Automated tests	Coverage
RNF-SEG-01	Each access granted to a user must last 15 minutes and renew automatically while the session remains valid	JwtTokenIssuerTest, RefreshSessionServiceTest	Automated
RNF-SEG-02	If an attempt is made to reuse a renewal credential that was already used, the system must invalidate the…	RefreshTokenTest, RefreshSessionServiceTest, RefreshLifecycleIntegrationTest	Automated
RNF-SEG-03	The system must only serve requests from authorized web applications	SecurityCorsConfigurationTest, SecurityCorsWebTest	Automated
RNF-SEG-04	The data that measuring devices send must travel encrypted	—	No coverage
RNF-SEG-05	Recovery codes must be stored in a protected way, so that a data leak does not allow them to be used	PepperedHmacResetCodeHasherTest	Automated
RNF-SEG-06	Redeeming recovery codes must be limited to 10 attempts per address per 15-minute window, and tell the…	InMemoryResetAttemptLimiterTest, ResetAttemptLimiterBeforeDatabaseTest, PoolSaturatedLimiterTest, AuthControllerWebTest	Automated
RNF-SEG-07	The internal functions used to authenticate devices must only be reachable from the system's own network…	—	No coverage
RNF-SEG-08	The system must protect user information in accordance with Law 1581 of 2012 (Congreso de la República de…	—	No coverage
RNF-SEG-09	Device credentials must be checked securely	ApiKeyHashTest, DeviceApiKeyStorageIntegrationTest, BrokerAccessServiceTest	Automated
RNF-SEG-10	The system's passwords and keys must not be exposed in the project's shared files	—	No coverage

Table 25

Traceability of the RNF-REN requirements

Requirement	Description	Automated tests	Coverage
RNF-REN-01	Data queries and charts must respond in under 2 seconds	—	No coverage
RNF-REN-02	The system must process each device's information at a cadence of one reading every 60 seconds	—	No coverage
RNF-REN-03	Sending emails must not keep the user waiting more than 5 seconds	—	No coverage

Table 26

Traceability of the RNF-DIS requirements

Requirement	Description	Automated tests	Coverage
RNF-DIS-01	No target availability percentage is set for this version	—	No coverage
RNF-DIS-02	The system must tolerate the occasional loss of readings	DeviceConnectivityServiceTest	Automated
RNF-DIS-03	If the system is unavailable, the device must keep measuring and hold up to 30 samples (30 minutes)	—	No coverage
RNF-DIS-04	The service that receives the devices' data must retry its connection to the system every 10 seconds while…	—	No coverage

Requirement	Description	Automated tests	Coverage
RNF-DIS-05	If there is no Wi-Fi for 10 minutes, the device must restart	—	No coverage
RNF-DIS-06	The system must allow its health status to be checked	—	No coverage

Table 27

Traceability of the RNF-COM requirements

Requirement	Description	Automated tests	Coverage
RNF-COM-01	The web application must work in current browsers (Chrome, Firefox, Safari and Edge)	—	No coverage
RNF-COM-02	Linking over Bluetooth from the web is only available in desktop Chrome and Edge and Chrome on Android,…	—	No coverage
RNF-COM-03	The mobile application must run on Android devices	—	No coverage
RNF-COM-04	The measuring device must include a dedicated electrical measurement sensor, with one sensor per board,…	—	No coverage

Table 28

Traceability of the RNF-MAN requirements

Requirement	Description	Automated tests	Coverage
RNF-MAN-01	The system must be organized in functional modules (User and Profile Management, Home Management, Device…	ModularityTests	Automated
RNF-MAN-02	Every change to the data structure must be reversible	—	No coverage
RNF-MAN-03	Deleted data must be kept marked as deleted and not physically erased	HomePersistenceTest, UserPersonPersistenceTest, DeleteHomeServiceTest	Automated
RNF-MAN-04	The system must have automated tests that verify its main functions	—	No coverage

Table 29

Traceability of the RNF-I18 requirements

Requirement	Description	Automated tests	Coverage
RNF-I18-01	The web application must be available in Spanish, English, French and Portuguese	—	No coverage
RNF-I18-02	The mobile application must be available in Spanish, English, French and Portuguese	locales.test.ts (mobile)	Automated
RNF-I18-03	Alerts and recommendations must be shown in the language each user has chosen	—	No coverage

Table 30

Traceability of the RNF-ESC requirements

Requirement	Description	Automated tests	Coverage
RNF-ESC-01	The system must be able to grow in number of users and devices without affecting the user experience	—	No coverage
RNF-ESC-02	No target numbers of concurrent users and devices are set for this version	—	No coverage

Table 31

Traceability of the RNF-USA requirements

Requirement	Description	Automated tests	Coverage
RNF-USA-01	The interface must be clear, adaptable to different screen sizes and consistent, with immediate feedback	—	No coverage

Table 32

Traceability of the RNF-INT requirements

Requirement	Description	Automated tests	Coverage
RNF-INT-01	Measuring devices must communicate with the system through connections started from the home, so that they…	—	No coverage
RNF-INT-02	The system must send emails through an email account configured for that purpose	EmailSenderTest	Automated
RNF-INT-03	The system must allow access with Google accounts, requesting only the profile name, email and photo	SignInWithGoogleServiceTest	Automated
RNF-INT-04	Browser notifications must only work on a secure connection and require the system to have that service…	—	No coverage
RNF-INT-05	Device configuration over Bluetooth must be available for up to 10 minutes, must not interrupt measuring…	—	No coverage

Requirements with no coverage are known gaps (see risks and limitations). Account administration has no requirement in the specification (DOC-01); the specification now covers it with the RF-ADM requirements.

Results

Full backend suite with MySQL (version 1.1)

On October 20, 2026 the full core service suite (`./mvnw test`, test profile, MySQL 8.4) finished with **1131 tests, 0 failures and 0 errors**. The required `Tests` check of GitHub Actions gives the same result on `main` (commit `90ae5ef`, 23 min 9 s), and it runs on every pull request to `dev` and `main`. The per-module distribution of the most recent local Surefire report (October 16, 2026: 132 classes, 1124 tests, 0 failures) is in full-suite-summary.md; 7 tests were added between that report and the final run.

Suites per repository (version 1.2, 10/24/2026)

With the services architecture, each repository runs its own suite with `./mvnw test` (against MySQL 8.4 for those that need it) locally and in the `Tests` check of its pipeline. Results, all green:

- `energy-monitor-backend` (home, alert, recommendation, notification and intermediaries): 481 tests; `ModularityTests` green.
- `en-monitor-measurement-service`: 151 tests.
- `en-monitor-device-service`: 130 tests.
- `en-monitor-api-gateway`: 57 tests.
- `en-monitor-security-service`: 660 tests (run of October 22, 2026).

End-to-end tests on the full local stack (`energy-monitor-infra`, with the five services `healthy`):

- `e2e/device-flow.sh` (21/21): gateway routes, linking a module (the device service queries the backend), broker authentication against the device service with a correct and a wrong key, telemetry (`ONLINE` and the reading stored by the measurement service), `offline` status (`device.connectivity-lost` event and `CONNECTIVITY` alert in the backend) and home deletion (`home.deleted`, the module is unlinked and receives `unlink`).

- `e2e/measurement-flow.sh` (25/25): two readings published through the broker that the measurement service stores only once and announces through its *outbox* (`measurement.recorded`), a `THRESHOLD` alert raised and resolved in the backend, consumption, history, statistics and levels through the gateway, and the `energy-monitor-backend.measurement.recorded` queue and its `.dlq` empty.

No Surefire reports of version 1.2 are attached to the evidence.

Backend without a database (version 1.0 run, 10/14/2026)

`./mvnw -o test` was run limited to the classes that do not boot a context with a database (88 existing classes) plus the 4 new classes. The totals per module come from target/surefire-reports and are detailed in surefire-summary.md.

Table 33

Backend results per module

Module	Classes	Run	Passed	Failed	Skipped	Time (s)
alert	9	48	48	0	0	1.2
device	6	48	48	0	0	2.0
home	23	159	159	0	0	1.5
measurement	11	67	67	0	0	0.8
notification	8	60	60	0	0	1.4
recommendation	4	23	22	1	0	0.3
security	31	430	429	1	0	18.8
Total	92	835	833	2	0	26.0

37 classes that need MySQL (270 declared methods) were not run because the SPRING_DATASOURCE_PASSWORD variable was not defined in the execution environment. They are the persistence tests, the integration tests, the web tests with a full context, ModularityTests and the security tests with a context; their status is “To run (MySQL)”.

Web application

`npm test` was run: 20 files and 141 tests, all passed (vitest-summary.md). In version 1.1 (October 20, 2026, branch `main`, commit `2b41cb3`) `npx vitest run` repeated the result: 20 files and 141 tests passed. The `Tests and build` check of GitHub Actions runs these tests and the production build on every pull request.

New tests

4 classes were written for critical requirements with no automated coverage at all. They follow the style of the existing tests (@WebMvcTest with @MockitoBean, or plain Mockito) and the AAA structure.

Table 34

New tests and why they were written

Class	Tests	Passed	Failed	Reason	Requirements
BrokerAuthControllerTest	12	12	0	Device authentication and authorization at the broker (security); the controller had no test	RF-DEV-07, RNF-SEG-09
HomeDeviceControllerTest	10	10	0	Device linking and credentials; only the use case had a test	RF-DEV-02 a RF-DEV-06
RecommendationControllerTest	7	6	1	Recommendation inbox; the controller had no tests. In version 1.1, with DEF-02 closed, all 7 pass	RF-REC-08
TelemetryMqttConsumerTest	10	10	0	Reading ingestion and dropping of foreign messages; the MQTT consumer had no tests	RF-MED-02, RF-MED-03

Defects and observations

Table 35

Defects and observations found

ID	Severity	Item	Description	Status
DEF-01	Low	SecuritySupportModelTest.refusesMissingNames	The test requires Person to reject a null last name, but the code made it optional on purpose (changeset security-023: a Google account may have no last name). The test is out of date; the code is correct. It was not changed. Version 1.1: the test was updated; it requires the first name and accepts a missing last name, and it passes.	Closed
DEF-02	Medium	RecommendationControllerTest.listRequiresHomeId (prueba nueva)	A missing required parameter (for example homeId in GET /api/v1/recommendations or /api/v1/alerts) answers 500 instead of 400: SecurityExceptionHandler declares a globally scoped @ExceptionHandler(Exception.class) that catches MissingServletRequestParameterException. The test was left failing; it was not adjusted. Version 1.1: `handleUnexpected` keeps the 4xx status of framework exceptions (`ErrorResponse`) and `TypeMismatchException` answers 400; the test passes.	Closed
DOC-01	Low	Documentación	Account administration (routes /api/v1/admin, permissions USER.READ, USER.MANAGE and ROLE.MANAGE, the /admin/users screen and administrator notifications) exists in the code but has no requirement in the requirements specification and is not described in the technical and administrator manuals. Version 1.1: the technical and administrator manuals now describe it; the specification now includes the requirement (RF-ADM).	Resolved

Version 1.3 (10/28/2026)

With the complete migration each repository runs its suite with `./mvnw test` (against MySQL 8.4 for those that need it) and the web with `npx vitest run`. Table 36 gathers the results, all green.

Table 36

Suites per repository, version 1.3

Repository	Commit	Tests	Result
en-monitor-security-service	f57c76b	773	0 failures
en-monitor-home-service	f424083	260	0 failures
en-monitor-notification-service	7b10398	234	0 failures
en-monitor-measurement-service	8a9b544	151	0 failures
en-monitor-device-service	06c1bdc	129	0 failures
en-monitor-alert-service	f050b56	119	0 failures
en-monitor-recommendation-service	aeeeec5	71	0 failures
en-monitor-api-gateway	0b2c164	69	0 failures
energy-monitor-web	698d4c9	228 (38 files)	0 failures

End-to-end tests on the complete local stack (`energy-monitor-infra`, with the eight services `healthy`). Table 37 details what each one verifies.

Table 37

End-to-end tests of energy-monitor-infra

Script	Checks	What it covers
`e2e/device-flow.sh`	24	Gateway routes, linking a module, broker authentication with a correct and an incorrect key, telemetry, `offline` status and `CONNECTIVITY` alert, and home deletion with the `unlink` command
`e2e/measurement-flow.sh`	28	Readings stored once and announced by the *outbox*, `THRESHOLD` alert raised and resolved, consumption, history, statistics and levels through the gateway, and queues with no pending messages

Script	Checks	What it covers
`e2e/alert-flow.sh`	36	Consumption and connectivity alerts end to end across the measurement, device, alert and notification services
`e2e/activity-flow.sh`	60	Home and device activity through the single notification queue: welcome, members, thresholds, ownership and deletion, with its text per recipient
`e2e/account-deletion-flow.sh`	12	Account created with Google: it cannot change the password (409), asks for the code by email (security, *outbox*, notifications), a wrong code does not delete and the right one does

The `e2e/recommendation-flow.sh` script also exists and was not run again in this version. The gateway limits (20 POST per minute per IP and path) mean that consecutive runs of the same script may answer 429.

Mobile Application

The mobile application had no automated tests. In version 1.3 a unit suite was added and run, together with two static checks, on `dev` (commit `1e4109b` plus the test files, October 28, 2026), and a manual protocol was defined:

- Unit tests with Jest and `jest-expo` (`npm test`): **8 suites and 78 tests, all passed** (October 28, 2026). They cover the password rules, the registration schema, the verification code, the registration body (trimmed and with `acceptTerms`), the terms prompt (accept, cancel and dismiss), the normalization of network errors and of the 429 with `Retry-After`, the mapping of each operation and status to its message (428, 409, 429, 503), and the parity of the texts in the four languages, including that every error key the API can produce exists in all of them.
- `npx tsc --noEmit`: 0 errors.
- `npx expo lint`: no errors (exit code 0).

Unit tests do not replace a test on a device: they do not walk through screens nor make the real API call.

Table 38 is the manual test protocol for an Android device. Its status is **to be run**: none has been performed on a device; the mobile application requirements that become «Automated» are only those the unit suite covers (RF-CLI-05 and RNF-I18-02). Before running them a new APK must be built, because the previous one does not send `acceptTerms` and registration would answer 428.

Table 38

Manual test protocol of the mobile application

ID	Test	Expected result	Requirements	Status
MOV-01	Register without ticking the terms and then ticking them	Unticked it is not sent; ticked the code is requested and the verification screen opens	RF-SEC-01, RF-SEC-13	To be run
MOV-02	Register with a misspelled email domain (`name@meil.com`)	The registration is rejected with a readable message	RF-SEC-01	To be run
MOV-03	Verify the email with the received code and sign in	The account is created on verification and the dashboard opens	RF-SEC-02, RF-SEC-03	To be run
MOV-04	Sign in and out; reopen the application	The session is kept on reopening and closed on exit	RF-SEC-03, RF-SEC-04	To be run
MOV-05	Sign in with Google for the first time without accepting the terms	The terms are requested and, once accepted, the account is created	RF-SEC-05, RF-SEC-13	To be run
MOV-06	Recover the password (email, code and new password)	It is possible to sign in with the new password	RF-SEC-06	To be run
MOV-07	Create a home, mark it as a favorite and change its thresholds	The home appears in the list and in favorites with the new thresholds	RF-HOM-01, RF-CLI-04	To be run
MOV-08	View the dashboard of a home with a meter sending data	Consumption and the last reading are shown	RF-MED-07, RF-MED-08, RF-CLI-04	To be run
MOV-09	Receive an alert and open the notification inbox; change the preferences	The alert is listed, marked as read and the preferences are saved	RF-ALE-04, RF-NOT-01, RF-CLI-04	To be run

ID	Test	Expected result	Requirements	Status
MOV-10	Edit the first and last name separately; change the password	Each field is saved on its own; a Google account cannot change the password	RF-SEC-07, RF-SEC-08	To be run
MOV-11	Delete the account (with password; with Google, with the code from the email)	The account is deleted and the email can register again	RF-SEC-11	To be run
MOV-12	Change the language among the four available and turn on dark mode	Texts and colors change without restarting	RF-CLI-05, RF-CLI-06	To be run

Version 1.4 (10/28/2026)

With the latest changes the totals of five repositories change. Table 39 shows which tests were added and the `main` commit.

Table 39

Changes in the suites, version 1.4

Repository	Commit	Tests	What was added
en-monitor-security-service	de52c95	774	The internal profile includes the avatar (`UserProfileQueryServiceTest`)
en-monitor-home-service	6a4e396	260	The members return `profileImage` (adapter and endpoint, no new tests: the existing ones were extended)
en-monitor-notification-service	ecdbc0a	235	Wording of the unusual consumption alert (`AlertMessagesAnomalyTest`)
en-monitor-alert-service	0e65430	128	Unusual consumption rule (`RegisterAnomalyAlertServiceTest`, 8) and its place in the consumer (1)
energy-monitor-web	27ad9f0	253 (44 files)	Photo viewer, photos of the members, Settings help modals, homes list that closes on an outside click and push in Brave
energy-monitor-mobile	8b5edc0	78	No changes in the tests

The tests that need MySQL of the alert and security services were not run on the work machine; the `Tests` check of each pipeline runs them.

Evidence

The evidence is in evidence: Surefire summary, trimmed execution log and Vitest summary from version 1.0, the full suite summary and new Vitest run from version 1.1, and the status of version 1.2 (totals per repository and end-to-end tests). It contains no secrets.

Risks, Limitations and Known Gaps

Table 40

Risks and Limitations of the Tests

ID	Risk or limitation	Effect	Recommendation
R-01	Non-hermetic tests: persistence and full context need real MySQL and SPRING_DATASOURCE_PASSWORD	Mitigated in version 1.1: the full suite runs with MySQL locally and in CI; the tests are still not hermetic	Evaluate Testcontainers
R-02	The mobile application has no automated tests	Mitigated in version 1.3 with 78 unit tests, `tsc`, `expo lint` and a manual protocol still to be run	Run the protocol (Table 38) and add component and screen tests

ID	Risk or limitation	Effect	Recommendation
R-03	MQTT with QoS 0	A reading may be lost; loss is not tested	Evaluate QoS 1 and add a loss test
R-04	No over-the-air (OTA) update	Firmware is updated only by USB	Document the USB upload procedure
R-05	Web Bluetooth limited to Chrome, Edge and Chrome for Android, and to HTTPS or localhost	Not verified automatically, nor is the EnergyMonitor-Setup portal	Evaluate browser end-to-end tests
R-06	The recovery attempt counter lives in memory	It resets with the service and is not shared between instances	Shared limiter if scaled
R-07	No performance or accessibility tests	RNF-REN and RNF-USA-01 without coverage	Define the targets and measure them
R-08	No coverage report (JaCoCo is not in pom.xml)	Coverage percentage unknown	Add JaCoCo (not added without asking)
R-09	Gaps with no automated test: StatusMqttConsumer, DeviceInactivityScheduler, HomeDeletedListener, DeviceCommandListener, MqttDeviceCommandAdapter, RecommendationScheduler, NotificationChannelsController, HomeConsumptionController, ApplianceTypeController, RegisterConnectivityAlertService, DeviceConnectivityLostListener, BrokerAuthSecurityConfiguration	No direct automated coverage	Prioritize connectivity and disconnection alerts

| R-10 | The full backend suite takes about 23 minutes in GitHub Actions | Pull requests wait that long before they can be merged | Parallelize the suite or split out the slowest integration tests |

Conclusions and Recommendations

The core service has a broad and fast suite: 835 tests run without a database in 26 seconds, with 2 failures that are not production code (DEF-01) or that point to an error-handling defect (DEF-02). The web passes its 141 tests. The 37 classes with MySQL remain to be run to close the verification. Recommendations: (1) define SPRING_DATASOURCE_PASSWORD and run the full `./mvnw clean test`; (2) fix DEF-02 with a MissingServletRequestParameterException handler; (3) update the DEF-01 test; (4) adopt Testcontainers to make the persistence tests hermetic; (5) add tests to the mobile application; (6) add JaCoCo and continuous integration; (7) write the administration requirements and their tests.

Version 1.1: the verification of the core service is closed with the full suite (1131 tests, 0 failures) against MySQL, locally and in GitHub Actions. Recommendations (1), (2) and (3) were carried out, and the continuous integration of (6) now exists with required checks on `dev` and `main`. Testcontainers (4), mobile application tests (5), JaCoCo (6) and the administration requirement in the specification (7) remain pending.

Version 1.2: with the services architecture, verification is split between the repositories and every suite is green (backend 481, measurements 151, devices 130, gateway 57 and security 660), with `ModularityTests` green and the two end-to-end tests of `energy-monitor-infra` passing (21/21 and 25/25) on the local stack. The same recommendations (4) to (7) remain pending, now in each Java service; JaCoCo is in none of the `pom.xml` files. Redoing the per-class inventory and the traceability per repository is pending.

Version 1.3: with the complete migration, the eight Java suites and the web are green (773, 260, 234, 151, 129, 119, 71, 69 and 228 tests) and the five end-to-end tests pass. The traceability incorporates the acceptance of terms (RF-SEC-13) and the rules for Google accounts. The biggest gap is still the mobile application: it has 78 unit tests, but none of components or screens, and its manual protocol has not been run. Recommendations: run that protocol on a device, add component tests to the mobile application (React Native Testing Library), redo the per-class inventory of the services and run `recommendation-flow.sh` again.

References

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